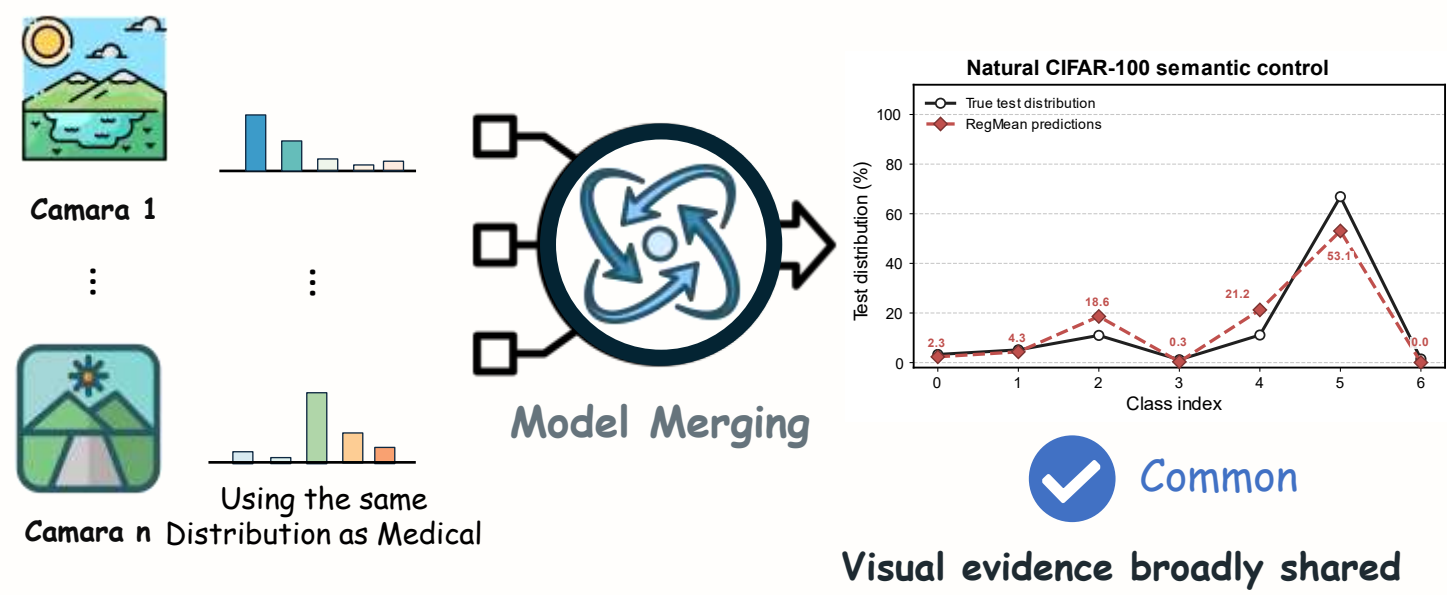
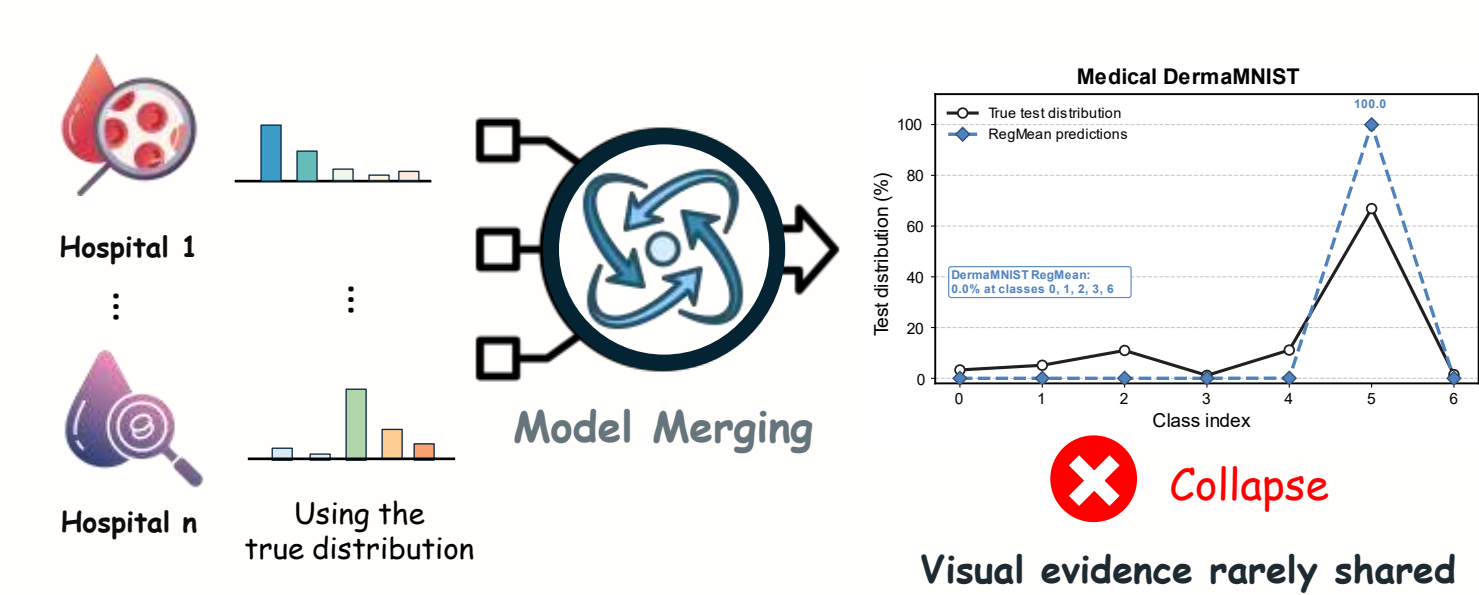


(a) Natural Image Merging



(b) Medical Image Merging



Observation 1: Diagnostic evidence is not shared

Each hospital learns non-overlapping diagnostic features

Hospital 1: Only class A, B available
Hospital n: Only class C available

Problem 1: Prediction collapses

Most test images are assigned to one dominant diagnosis

Collapsed!

Observation 2: Long-tailed distribution

Medical test set is naturally long-tailed: common diagnoses dominate.

True label distribution

Problem 2: Model collapses, Accuracy high

High raw ACC but poor multi-class discrimination

90%
X